Answers

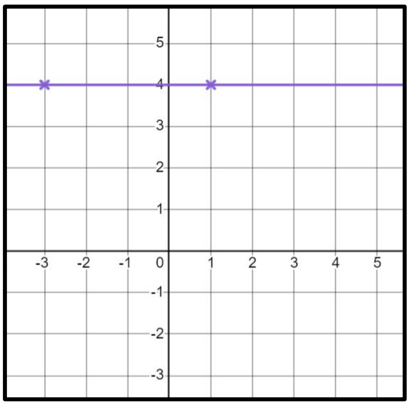
Worksheet

Plotting a relationship

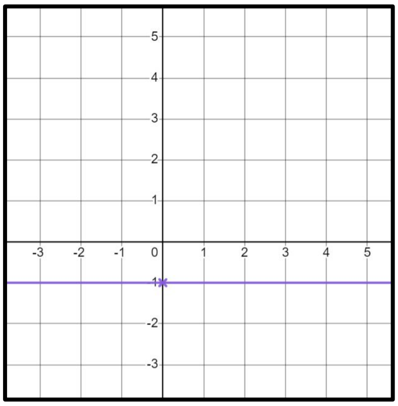
Task A: Plotting lines from coordinates

1) For each rule, fill in the coordinates. Use them to draw the line for that rule.

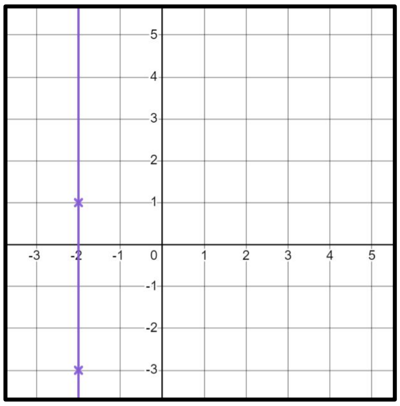
a) $y = 4$
(1,4) (-3,4) E.g. (3,4)



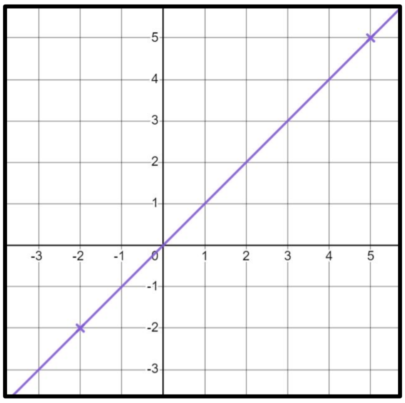
b) $y = -1$
(0,-1) (?,-1) E.g. (4,-1)



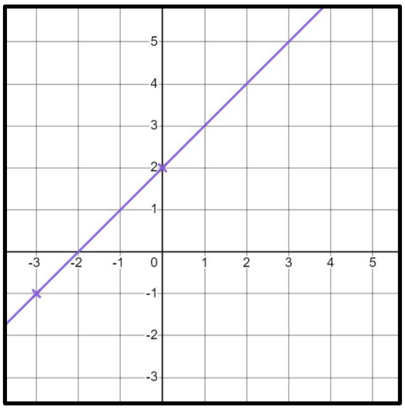
c) $x = -2$
(-2,-3) (-2,1) E.g. (-2,4)



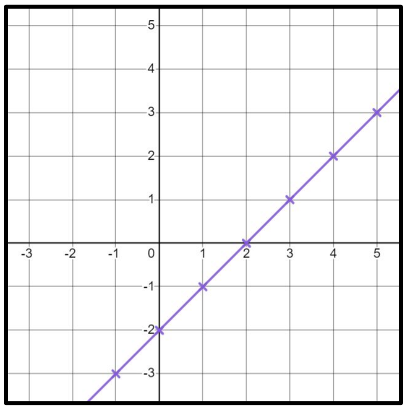
d) $y = x$
(-2,-2) (5,5) E.g. (1,1)



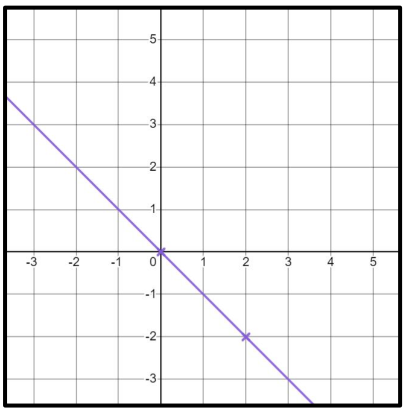
e) $y = x + 2$
(-3,-1) (0,2) E.g. (3,5)



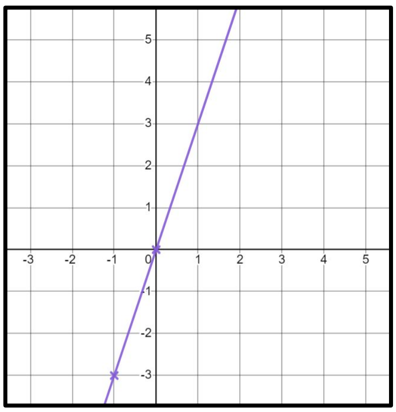
f) $y = x - 2$
E.g.(-1,-3) (0,-2) (1,-1) (2,0) (3,1) (4,2) (5,3)



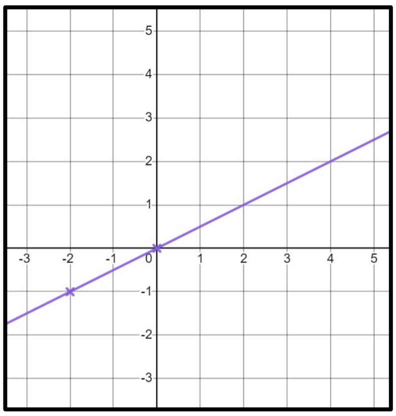
g) $y = -x$
(2,-2) (0,0) E.g. (-3,3)



h) $y = 3x$
(0,0) E.g. (-1,-3) (1,3)



i) $y = \frac{1}{2}x$
(0,0) (-2,-1) E.g. (4,2)



Task B: Identifying whether points are on a line

1) For each equation, does the coordinate lie on the given line?
Give a reason for your answer.

a) (4,18) $y = 5x$ No $4 \times 5 \neq 18$

b) (6,-12) $y = -2x$ Yes $6 \times -2 = -12$

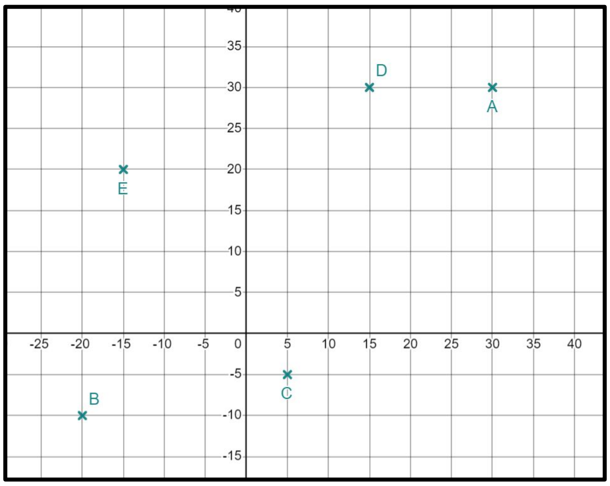
c) (0,0) $y = 3.5x$ Yes $0 \times 3.5 = 0$

d) (25,22.5) $y = x - 5.5$ No $25 - 5.5 \neq 22.5$

e) (11,37) $y = 3x + 4$ Yes $11 \times 3 + 4 = 37$

f) (10, 50) $y = 5x - 8$ No $10 \times 5 - 8 \neq 50$

2) Alex has started to plot some lines.
He has plotted one point from each line but cannot remember which point goes with which line.
Match the points with the equation for the line Alex has tried to plot.



a) $y = 20$ E (-15,20)

b) $y = x$ A (30,30)

c) $y = 2x$ D (15,30)

d) $y = x - 10$ C (5,-5)

e) $y = x + 10$ B (-20,-10)